

Be a SPARK. Take the LEAD.

MODULE No.: 2

TOPICS COVERED: PYTHON CONDITIONAL STATEMENTS

TYPE: [] WW [/] ST [] PT [] TE [] FOR

NO. 2

TOTAL SCORE

20

NAME: _____

SECTION: _____

TEACHER: _____

DATE: _____

Directions:

1. Read all instructions carefully before starting the assessment.
2. Choose **ONE (1)** challenge only:

Option 1: Electricity Consumption Checker

Option 2: Flood Warning System
3. Create your Python program in **Google Colaboratory**.
4. Follow the required **variable names**, **data types**, and **conditions** provided in your chosen option.
5. Your program must:

Use variables.

Use the correct data types (str and int).

Collect user input using input().

Display output using print().

Use an if/else conditional statement.

Display the appropriate recommendation or warning message based on the user's input.
6. Test your program to ensure it runs correctly and produces the expected output.
7. Rename your Google Colaboratory file using the format: **T1.ST2_FULL NAME**
8. Verify that the correct file has been uploaded before clicking **Submit Assignment**.

Option 1: Electricity Consumption Checker (SDG 7: Affordable and Clean Energy)

Scenario: Electricity is an important resource used every day at home and in school. Monitoring electricity consumption can help people save energy and reduce costs. Create a Python program that helps users evaluate their daily electricity consumption and displays an appropriate recommendation.

Required Variables and Data Types

Variable Name	Data Type	Purpose
name	str	Stores the user's name
rainfall_level	int	Stores the electricity consumed in kilowatt-hours

Program Requirements

1. Ask the user to enter their name.
2. Ask the user to enter the amount of electricity consumed today in kilowatt-hours (kWh).
3. Store the inputs using the required variable names.
4. Use an if/else statement to check the value of electricity_kwh.
5. Display:

- "Consider reducing your electricity usage." if electricity_kwh > 10
 - "Great job conserving energy!" otherwise
6. Display the user's name and recommendation.
 7. Add at least one code comment explaining the condition.
 8. Test the program using the provided test cases.

Option 2: Flood Warning System (SDG 11: Sustainable Cities and Communities)

Scenario: Heavy rainfall can cause flooding in communities, putting people, homes, and property at risk. Technology can help people monitor rainfall levels and provide early warning messages to support preparedness and safety. Create a Python program that helps users determine whether a flood warning should be displayed based on the recorded rainfall level.

Required Variables and Data Types

Variable Name	Data Type	Purpose
name	str	Stores the user's name
rainfall_level	int	Stores the rainfall level in millimeters

Program Requirements

1. Ask the user to enter their name.
2. Ask the user to enter the recorded rainfall level in millimeters (mm).
3. Store the inputs using the required variable names.
4. Use an if/else statement to check the value of rainfall_level.
5. Display:
 - "Flood Warning: Stay Alert and Prepare for Possible Flooding." if rainfall_level > 50
 - "No Flood Warning: Conditions are Normal." otherwise
6. Display the user's name and warning message.
7. Add at least one code comment explaining the condition.
8. Test the program using the provided test cases.

Submission Instructions

1. Test your program to ensure it runs correctly and produces the expected output.
2. Rename your Google Colaboratory file using the format: **T1.ST2_FULL NAME**
Example: T1.ST2_Juan_Dela_Cruz.ipynb
3. Upload the project file to the submission area in this Canvas assignment.
4. Verify that the correct file has been uploaded before clicking **Submit Assignment**.

Academic Integrity Reminder

Complete this assessment independently. The use of AI tools (e.g., ChatGPT, Copilot, Gemini, or similar tools) to generate or complete your code is **not allowed**. All submitted work must reflect your own understanding and coding skills.

Rubric

Criteria	Advancing (4)	Benchmarking (3)	Connecting (2)	Emerging (1)	Points
Variables and Data Types	Correctly uses name (str) and electricity_kwh (int) OR rainfall_level (int) . All variables are declared, assigned, and used correctly throughout the program.	Uses the required variables and data types with one minor error in naming or usage.	Uses some required variables or data types, but multiple errors are present.	Variables and/or data types are mostly incorrect, missing, or not used appropriately.	
Input Statements	Correctly uses input() to collect the user's name and electricity consumption (kWh) OR rainfall level (mm) and stores them in the correct variables.	Input statements are present with one minor error in variable assignment or data conversion.	Input statements are partially correct but contain multiple errors.	Input statements are missing or incorrectly implemented.	
Conditional Statement (if/else)	Correctly uses an if/else statement with the required condition (electricity_kwh > 10 OR rainfall_level > 50) and displays the appropriate message.	Condition or decision structure contains one minor error but generally works as intended.	Condition is partially correct but contains significant errors affecting the output.	Conditional statement is missing or incorrect.	
Output Statements	Correctly displays the user's name and the appropriate recommendation/warning message using print() statements.	Output is mostly correct with minor formatting or display errors.	Output is incomplete or missing important information.	Output is incorrect, unclear, or missing.	
Program Execution and Accuracy	Program runs successfully without errors and produces the correct output for different input values.	Program runs with minor issues but generally produces correct output.	Program runs but contains several errors that affect accuracy.	Program does not run successfully or produces incorrect output.	
Total					/20